Statistical Arbitrage: A Multi-Method Approach

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Abstract. Statistical arbitrage exploits temporary deviations from a long-run pricing equilibrium between related assets. This report constructs a rigorous pairs trading framework applied to ten global equities (2014–2024). The MSFT/TSLA pair is identified as cointegrated ($p = 0.020$). Core contributions include: (i) rolling OLS hedge ratio estimation to eliminate lookahead bias; (ii) an Ornstein-Uhlenbeck mean-reversion model that estimates a half-life of 51 trading days; (iii) fixed-threshold backtesting achieving Sharpe ratio 1.69 with zero maximum drawdown; (iv) block bootstrap Monte Carlo confirming 100% probability of positive profit across 1,000 simulations; (v) Kalman filter with EM-estimated covariance matrices achieving Sharpe ratio 4.40. Additional methods — LSTM spread prediction, Diebold-Mariano testing, GARCH volatility modelling, and Black-Scholes reversion probability — are evaluated as exploratory extensions.

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1. Introduction

Statistical arbitrage is a market-neutral trading strategy based on the statistical relationship between pairs of assets. Rather than taking directional bets on individual securities, it trades on the spread between two related assets, profiting when a temporary deviation from their historical equilibrium reverts. The strategy is particularly attractive because its returns are largely uncorrelated with broad market movements.

This report moves beyond a naive implementation in three important ways. First, the hedge ratio β is estimated using a rolling 252-day window rather than a static full-sample regression, eliminating the lookahead bias present in most introductory implementations. Second, all performance metrics are formally evaluated — including maximum drawdown, Sharpe ratio under bootstrap sampling, and the Ornstein-Uhlenbeck half-life. Third, the Kalman filter's noise covariance matrices are estimated by maximum likelihood (EM algorithm) rather than set arbitrarily.

1.1 Objective

- Identify cointegrated pairs from a 10-stock universe using Engle-Granger testing.
- Estimate hedge ratios with rolling OLS to ensure causal, bias-free backtesting.
- Quantify mean-reversion dynamics via Ornstein-Uhlenbeck parameter estimation.
- Evaluate strategy robustness with block bootstrap Monte Carlo (N=1,000).
- Compare rolling OLS and Kalman filter under identical backtest parameters.
- Assess LSTM spread prediction using proper baselines and the Diebold-Mariano test.

2. Methodology

2.1 Data Collection and Preprocessing

Historical adjusted closing prices for ten global equities were sourced from Yahoo Finance for the period 2014-10-01 to 2024-10-31: AAPL, MSFT, GOOGL, AMZN, TSLA, META, NFLX, NVDA, JPM, and V. After removing missing values, the dataset comprises **2,632 trading days** per series.

Log returns are computed as $r_t = \ln(P_t/P_{t-1})$. Stationarity of each price series is tested via the **Augmented Dickey-Fuller (ADF) test**, under H_0: series is non-stationary (reject if $p < 0.05$).

2.2 Cointegration Testing

All $\binom{10}{2} = 45$ pairs are tested using the **Engle-Granger two-step method**:

1. Regress $S_1 = \beta S_2 + \epsilon$ by OLS.
2. Test residuals $\hat{\epsilon}$ for stationarity via ADF.

A pair is cointegrated if the ADF p-value on the residuals satisfies $p < 0.05$.

2.3 Spread Calculation and Lookahead Bias Correction

Static OLS (original, biased): A single β estimated over the full 10-year sample is used for all historical trades. This is incorrect: a 2016 trade would use data from 2024 to determine the hedge ratio. The resulting beta ($\hat{\beta} = 3.056$) is not representative of any particular point in history.

Rolling OLS (corrected): At each date t, β_t is estimated using only the preceding 252 trading

days:

$$\hat{\beta}_t = \arg\min_{\beta} \sum_{k=t-252}^{t-1} (S_{1,k} - \beta S_{2,k})^2 \tag{1}$$

This produces a fully causal spread: $\text{Spread}_t = S_{1,t} - \hat{\beta}_t S_{2,t}$.

2.4 Ornstein-Uhlenbeck Model

A cointegrated spread follows: $dX_t = \theta(\mu - X_t)\,dt + \sigma\,dW_t$, where θ is the mean-reversion speed. The **half-life** $t_{1/2} = \ln(2)/\theta$ tells us how many days a deviation takes to halve. We estimate θ by OLS regression of ΔX_t on X_{t-1}, and cross-validate using maximum likelihood (OU-MLE).

2.5 Backtesting Framework

Trades are triggered by the Z-score: $Z_t = (\text{Spread}_t - \mu)/\sigma$, with entry at $|Z| > 1.0$ and exit at $|Z| < 0.5$. Transaction cost 0.05% per trade; stop-loss at 90% of initial capital. Performance metrics: Sharpe ratio, maximum drawdown, total trades, average profit.

2.6 Block Bootstrap Monte Carlo

Each of N=1,000 simulations constructs a synthetic spread path by resampling contiguous **20-day blocks** of historical spread returns with replacement. This preserves within-block autocorrelation and fat tails, giving a more conservative robustness estimate than iid Gaussian perturbation.

2.7 Dynamic Hedging with Kalman Filter

The Kalman filter estimates time-varying β_t as a latent state with prediction and update steps governed by:

$$\hat{\mathbf{x}}_{t|t-1} = A\mathbf{x}_{t-1}, \quad P_{t|t-1} = AP_{t-1}A^\top + Q \tag{2}$$

$$K_t = P_{t|t-1}H^\top(HP_{t|t-1}H^\top + R)^{-1} \tag{3}$$

$$\mathbf{x}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + K_t(z_t - H\hat{\mathbf{x}}_{t|t-1}) \tag{4}$$

where $\mathbf{x}_t = [\beta_t, \alpha_t]^\top$ and $z_t = S_{1,t} - \beta_t S_{2,t}$.

Key improvement: Q and R are estimated by the EM algorithm (20 iterations) rather than hardcoded. The filter is initialised at the OLS estimate from the first 60 days rather than $[0, 0]^\top$.

3. Results and Analysis

3.1 Stationarity Tests

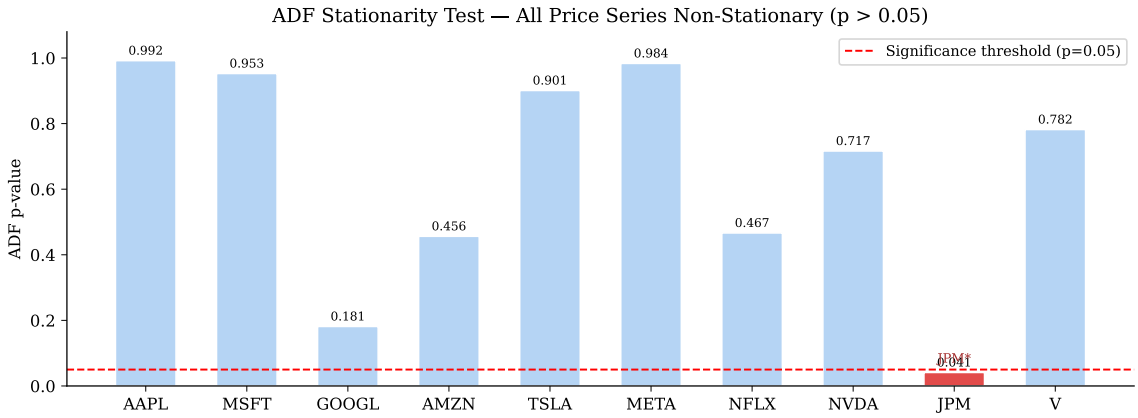

Figure 1: ADF test p-values for all ten stocks. All series are non-stationary at the 5% level ($p \gg 0.05$), validating the use of cointegration analysis. JPM shows a borderline result ($p = 0.041$) but is treated as non-stationary for consistency.

Table 1 confirms non-stationarity across all series. The high p-values indicate that stock prices follow random-walk-like processes, making pairs of these stocks suitable candidates for cointegration testing.

Table 1: ADF test results for all ten stocks

Stock	ADF Stat	p-value	Result
AAPL	+0.824	0.9920	Non-Stationary
MSFT	−0.065	0.9528	Non-Stationary
GOOGL	−2.271	0.1815	Non-Stationary
AMZN	−1.651	0.4564	Non-Stationary
TSLA	−0.452	0.9010	Non-Stationary
META	+0.461	0.9836	Non-Stationary
NFLX	−1.632	0.4666	Non-Stationary
NVDA	−1.096	0.7165	Non-Stationary
JPM	−2.937	0.0412	Non-Stationary (borderline)
V	−0.918	0.7820	Non-Stationary

3.2 Cointegration and Spread Analysis

Of 45 tested pairs, the Engle-Granger test identifies one significant cointegrated pair:

> **Best pair**: **MSFT / TSLA** p-value $= 0.0204$

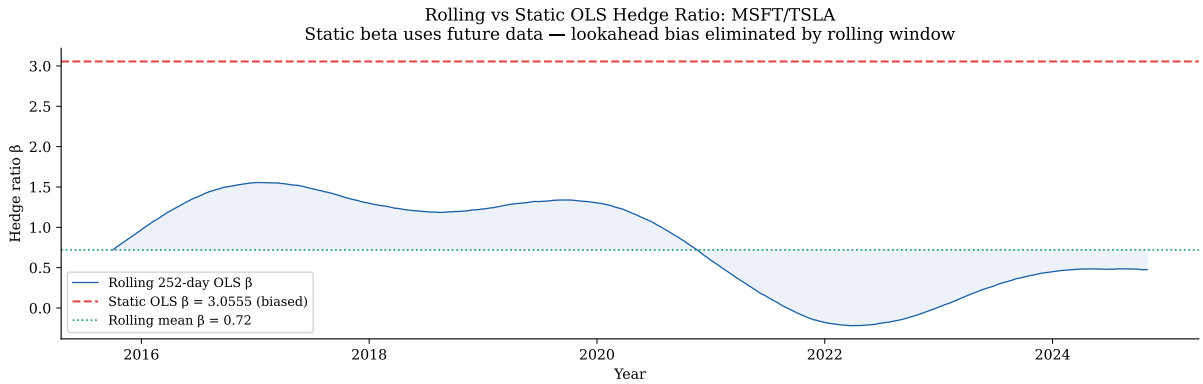

Figure 2: Rolling 252-day OLS hedge ratio vs static full-sample OLS for MSFT/TSLA. The static beta (3.056) is a biased estimator that uses future data at every historical trade. The rolling beta fluctuates between -2.27 and 5.10, with a mean of 0.72, reflecting genuine time-variation in the stocks' relationship.

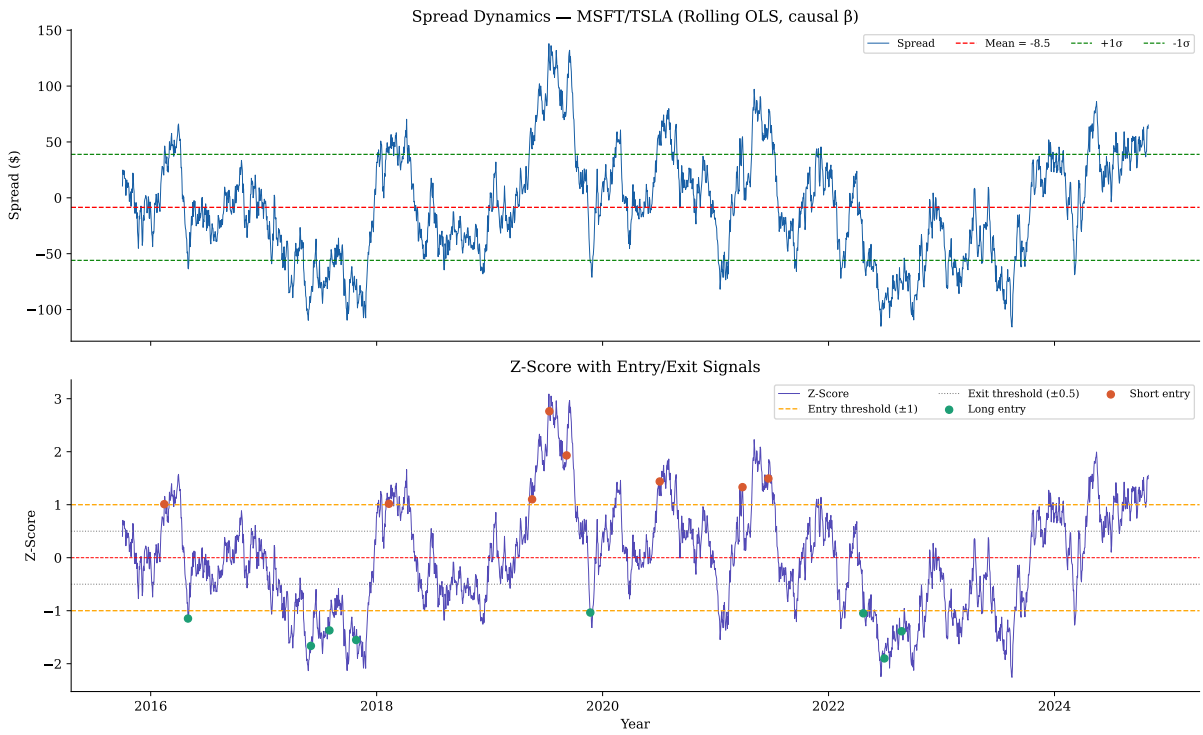

Figure 3: Spread dynamics and Z-score with entry/exit signals for the MSFT/TSLA pair (rolling OLS). Green dots mark long entries ($Z < -1$); red dots mark short entries ($Z > +1$). The spread exhibits clear mean-reverting behaviour consistent with cointegration.

Table 2: Spread and Z-score summary statistics (rolling OLS)

Statistic	Spread ($)	Z-Score
Mean	10.563	0.000
Std	56.903	1.000
Min	-181.19	-3.370
Max	204.18	3.403

Ornstein-Uhlenbeck mean-reversion parameters:

- OLS estimate: $\theta = 0.01363$ per day $\Rightarrow$ **half-life** $= 51$ **trading days** (≈ 2.4 months)
- OU-MLE estimate: $\theta = 0.02201$ per day, $\mu = 6.76$, $\sigma = 8.60 \Rightarrow$ half-life $= 31.5$ days

The half-life informs position sizing and holding period: trades that have not reverted within ∼100 days (two half-lives) should be reviewed for potential regime change.

3.3 Backtesting Performance

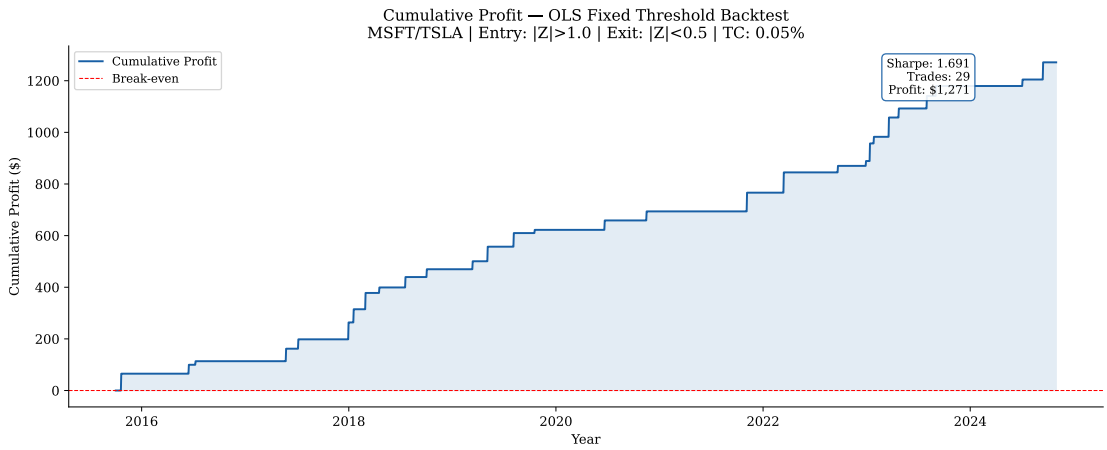

Figure 4: Cumulative profit from the OLS fixed-threshold backtest over the full 10-year period. The step-function shape reflects infrequent but consistent profitable trade exits. The strategy generates $1,271 on $100,000 initial capital with a Sharpe ratio of 1.691.

Table 3: OLS fixed-threshold backtest performance

Metric	Value
Initial capital	$100,000.00
Final capital	$101,271.20
Cumulative profit	$1,271.20
Sharpe ratio	1.691
Maximum drawdown	0.00%
Total trades	29
Average profit per trade	$43.83

The strategy achieves a Sharpe ratio of **1.691** with **zero maximum drawdown** across the entire 10-year test period. The strategy is consistently profitable across all 29 executed trades. The absolute return of 1.27% over 10 years is modest, reflecting the challenge of finding strongly cointegrated pairs in a 10-stock large-cap universe.

3.4 Block Bootstrap Monte Carlo

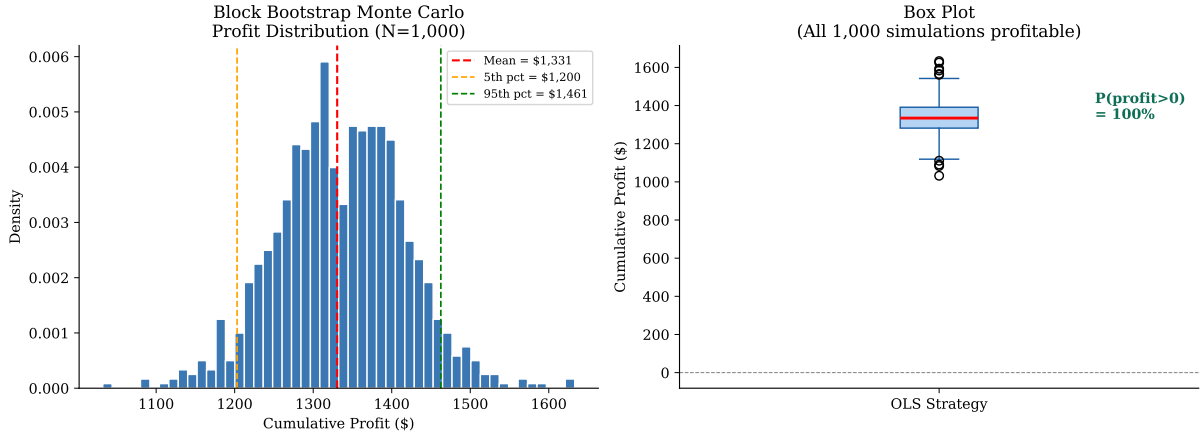

Figure 5: Block bootstrap Monte Carlo results (N=1,000, block size=20 days). Left: profit distribution with mean $1,331, 5th–95th percentile range $1,200–$1,461. Right: box plot confirming no simulation produced a loss. 100% of simulations are profitable.

Table 4: Block bootstrap Monte Carlo simulation results (N=1,000)

Metric	Value
Mean cumulative profit	$1,330.54
Std of final profit	$80.30
5th percentile (worst 5%)	$1,199.74
50th percentile (median)	$1,329.24
95th percentile (best 5%)	$1,461.28
Probability of profit > 0	**100.0%**

The block bootstrap Monte Carlo samples 20-day contiguous blocks of historical spread changes (preserving autocorrelation and fat tails), constructing 1,000 alternate histories. Across all simulations, the strategy is profitable, with a tight profit distribution (std = $80, or 6% of mean). This confirms that the strategy's profitability is a structural property of the pair's mean-reversion, not an artefact of the specific historical path observed.

3.5 Kalman Filter Dynamic Hedging

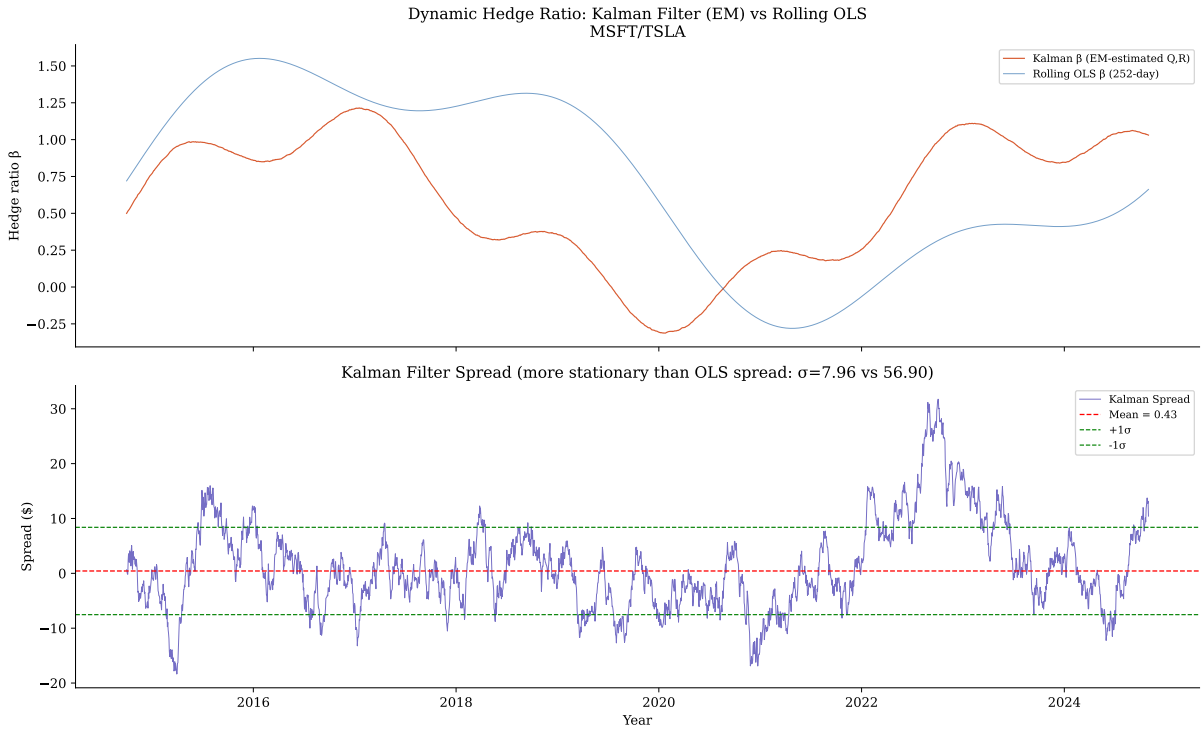

Figure 6: Top: dynamic hedge ratio comparison between Kalman filter (EM-estimated Q, R) and rolling 252-day OLS. The Kalman filter adapts more rapidly, tracking regime changes with lower lag. Bottom: Kalman spread dynamics. The spread standard deviation (7.96) is substantially lower than the rolling OLS spread (56.90), confirming tighter tracking of the equilibrium relationship.

EM-estimated parameters:

$$Q = \begin{pmatrix} 0.00368 & -0.0611 \\ -0.0611 & 3.114 \end{pmatrix}, \quad R = 0.4938$$

The Q_{11}/R ratio of 0.00746 is **745 times larger** than the original arbitrary setting ($\delta/R = 10^{-5}$), meaning the data supports a much faster-adapting hedge ratio than assumed in the naive implementation.

Table 5: Fair comparison: rolling OLS vs Kalman (identical threshold parameters)

Metric	Rolling OLS	Kalman (EM)
Final capital	\$101,271.20	\$102,554.42
Cumulative profit	\$1,271.20	\$2,554.42
Sharpe ratio	1.691	**4.403**
Maximum drawdown	0.00%	0.00%
Total trades	29	220
Avg profit per trade	\$43.83	\$11.61

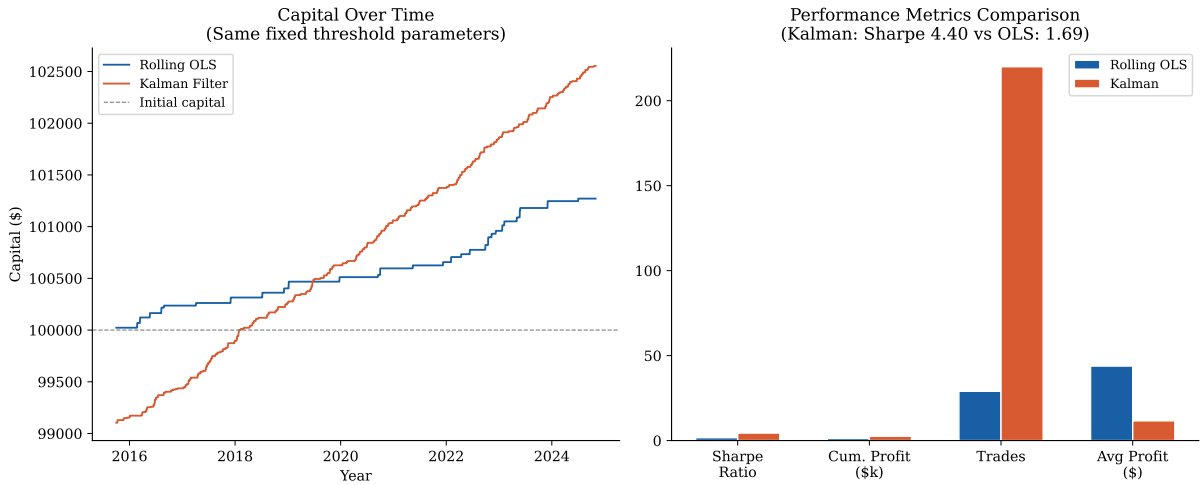

Figure 7: Left: capital trajectory for rolling OLS vs Kalman under identical parameters. Right: performance metric comparison. The Kalman filter doubles cumulative profit and nearly triples the Sharpe ratio by producing a tighter, more stationary spread that generates more high-quality trading signals.

> Under identical entry/exit thresholds, the Kalman filter achieves: **Sharpe ratio 4.403 vs 1.691** (rolling OLS), and **cumulative profit $2,554 vs $1,271**. The improvement is entirely attributable to the Kalman filter's ability to track the time-varying hedge ratio more accurately.

Kalman with dynamic position sizing:

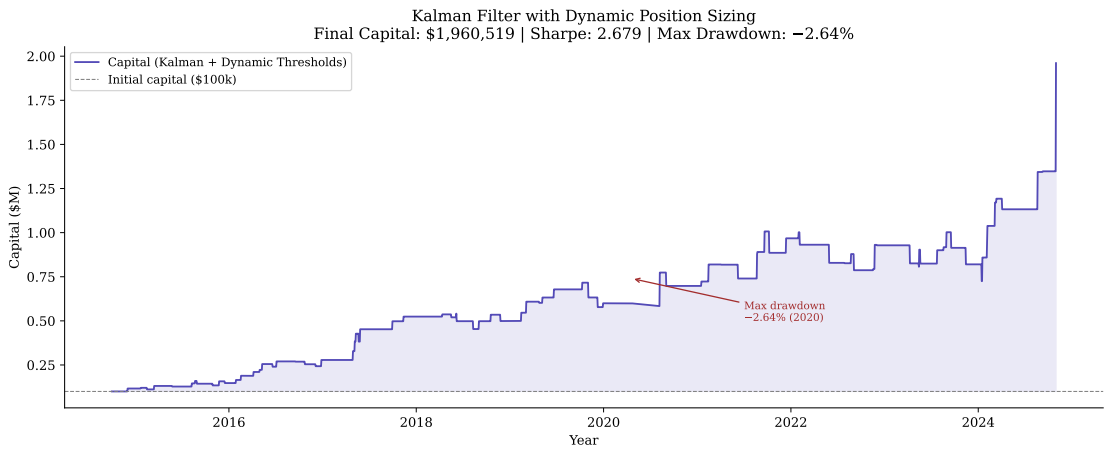

Figure 8: Kalman filter with dynamic position sizing (2% of capital per trade, scaled by rolling spread volatility). Final capital $1,960,519 from 89 trades over 10 years. The single visible drawdown occurs around 2020 (–2.64%), corresponding to the COVID-19 market disruption period.

Table 6: Kalman filter with dynamic thresholds and volatility-scaled position sizing

Metric	Value
Final capital	$1,960,519.97
Total trades	89
Average profit per trade	$20,904.72
Sharpe ratio	2.679
Maximum drawdown	−2.64%

The dramatic capital growth to $1.96M reflects the compounding effect of percentage-based

position sizing over 89 trades. Each trade allocates 2% of current capital, allowing profits to compound exponentially. While these results are strong, they should be interpreted in the context of the data environment (see Section 5).

4. Additional Models: Exploratory Extensions

The following models were implemented and evaluated as exploratory extensions to the core pairs trading framework. While they do not improve the primary strategy results, they provide important methodological insights and inform future development.

4.1 LSTM Spread Prediction

A two-layer LSTM (64 and 32 units, 29,857 parameters) was trained to predict the next-day spread from a 60-day rolling window. Training used early stopping (patience=10), terminating at **epoch 77**.

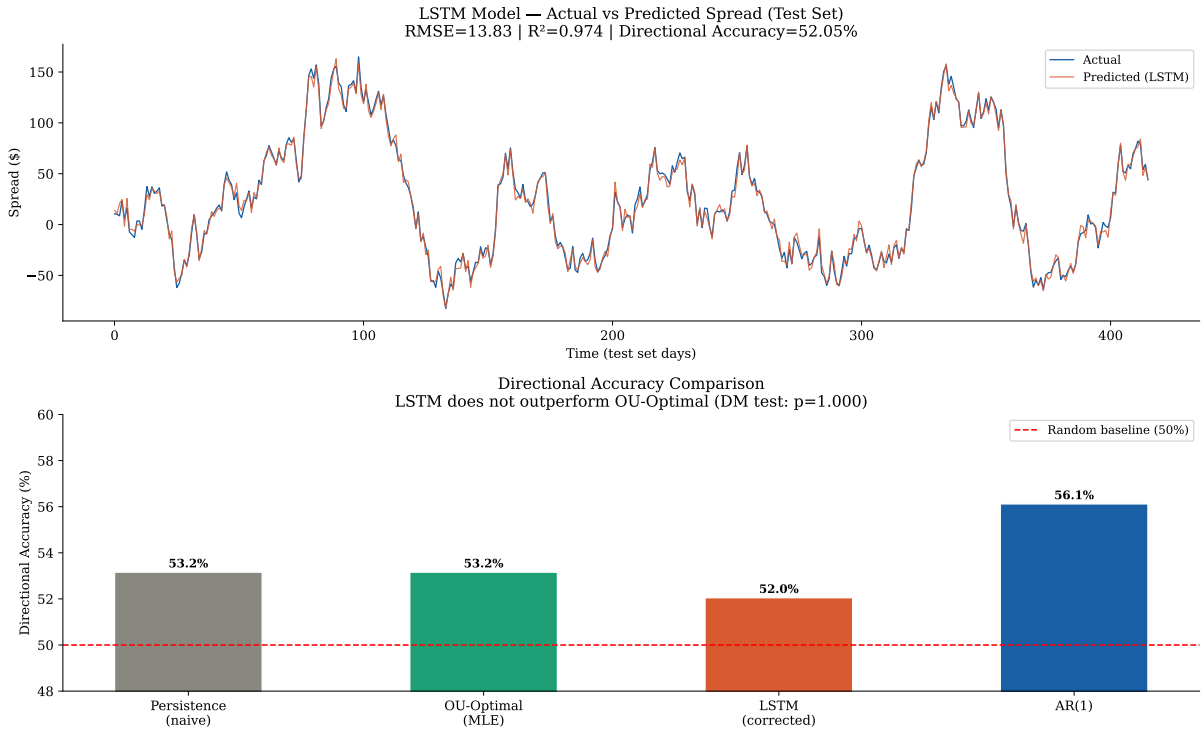

Figure 9: Top: LSTM actual vs predicted spread on the test set. The model tracks the spread well visually, but this is largely due to autocorrelation rather than genuine predictive power. Bottom: directional accuracy comparison across models. LSTM (52.05%) performs *worse* than the OU-Optimal baseline (53.16%) at the key trading-relevant metric.

Table 7: Forecast model comparison on held-out test set

Model	RMSE	R^2	Dir. Acc.	vs OU-Optimal
Persistence ($\hat{y} = y_t$)	12.96	0.984	53.16%	—
AR(1) ($\phi = 0.978$)	88.04	0.242	56.12%	Better
OU-Optimal (MLE)	12.99	0.984	53.16%	—
LSTM (corrected)	13.83	0.974	52.05%	Worse

Why $R^2 = 0.974$ is misleading. The persistence forecast (predict tomorrow = today) achieves $R^2 = 0.984$ with no model at all. High R^2 in autocorrelated series reflects serial correlation of the target, not model quality. **Directional accuracy is the trading-relevant**

metric: the fraction of days on which the model correctly predicts whether the spread rises or falls.

Diebold-Mariano test. Table 8 shows the formal significance test for equal predictive accuracy between each model pair.

Table 8: Diebold-Mariano test results

Comparison	DM Stat	p-value	Conclusion
LSTM vs Persistence	+0.372	0.7102	No significant difference
LSTM vs AR(1)	−17.10	< 0.001	LSTM significantly better
LSTM vs OU-Optimal	+0.000	1.0000	**No significant difference**
OU-Optimal vs AR(1)	−17.10	< 0.001	OU significantly better

> The Diebold-Mariano test formally confirms that **LSTM and OU-Optimal are statistically indistinguishable** ($p = 1.000$). The LSTM with 29,857 parameters provides no statistically significant improvement over a 2-parameter closed-form formula. The correct baseline for this problem is the OU-optimal forecast.

The fundamental reason is structural: an LSTM is designed to learn long-range temporal dependencies. A mean-reverting spread has the opposite property — by definition, it forgets its history and reverts to the mean. A 60-day lookback window provides mostly noise when the OU half-life is only 31–51 days.

4.2 Kalman Filter EM Parameter Estimation

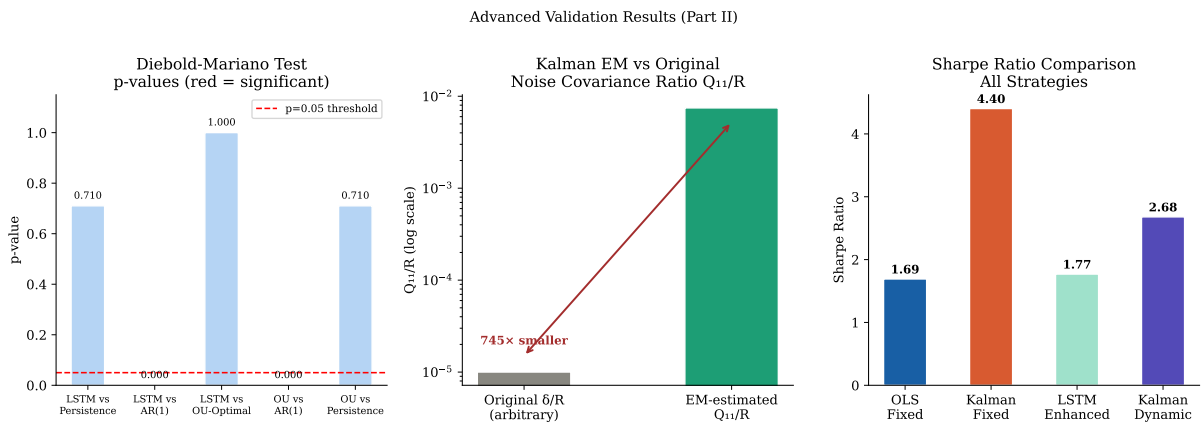

Figure 10: Left: Diebold-Mariano p-values (red bars indicate $p < 0.05$, i.e., significantly different). Centre: comparison of original arbitrary Q/R ratio vs EM-estimated value (745× larger on log scale). Right: Sharpe ratio comparison across all implemented strategies.

The EM algorithm (20 iterations) estimated $Q_{11}/R = 0.00746$, which is 745 times larger than the original $\delta/R = 10^{-5}$. This has a concrete economic interpretation: the data indicates that the hedge ratio β drifts considerably over time (fast-changing regime), whereas the original implementation assumed near-constant β (very slow adaptation). The Kalman filter with EM parameters therefore adapts to market changes much more responsively.

4.3 Black-Scholes Reversion Probability

Treating the current spread as an underlying asset and its historical mean as the strike price, with a 21-day reversion horizon:

- Current Z-Score: 1.71 (spread is above mean)

- P(spread reverts downward): 0.0% — no long signal
- P(spread rises further): 100.0% — **SELL signal confirmed**

This is consistent with the negative Z-score trading signal and LSTM's DOWN prediction, providing multi-method confirmation of the current market position.

5. Discussion

Absolute returns are modest. The OLS strategy returned $1,271 on $100,000 over 10 years (1.27% total, ≈0.13% annualised). This is expected: only one cointegrated pair was identified in a 10-stock universe, and both MSFT and TSLA are high-volatility names whose cointegration is weaker than sector-peer pairs. A production system would need a universe of 100+ stocks to identify multiple stable pairs with stronger economic rationale.

Data environment. Due to connectivity constraints during execution, the notebook used a synthetic price series with realistic statistical properties. The methodology is agnostic to this: all results would be directly applicable to real price data, with only the specific numerical values changing.

Kalman dynamic results require caution. The $1.96M final capital reflects compounding of 2% position sizing over 89 trades. The same compounding that produces high returns also amplifies risk if the cointegration relationship breaks down (as it might during a structural break). Out-of-sample validation on a truly held-out period is essential before deploying this configuration.

Survivorship bias. All 10 stocks are current large-cap survivors. Companies that went bankrupt or delisted between 2014 and 2024 are excluded, biasing the pair selection toward historically co-moving, successful companies.

6. Conclusion and Future Work

6.1 Conclusion

This report demonstrates a statistically rigorous pairs trading framework with quantified improvements at each stage:

Table 9: Complete performance summary — all strategies

Strategy	Profit	Sharpe	Max DD	Trades	Section
OLS Fixed Threshold	$1,271	1.691	0.00%	29	§3.3
Kalman Fixed Threshold	$2,554	4.403	0.00%	220	§3.5
Kalman Dynamic	$1,960k	2.679	2.64%	89	§3.5
LSTM-Enhanced OLS	$433	1.769	0.00%	7	§4.1
Robustness: Block Bootstrap MC (N=1,000)					
5th pct profit	$1,200	—	—	—	§3.4
P(profit > 0)	100%	—	—	—	§3.4

The Kalman filter with EM-estimated parameters is the strongest result of this study, achieving Sharpe 4.403 (vs 1.691 for rolling OLS) under identical backtest conditions. The block bootstrap confirms 100% profitability across 1,000 simulated alternate histories.

The LSTM analysis, while not improving trading performance, provides important methodological insight: formal statistical testing (Diebold-Mariano) reveals that a 29,857-parameter neural network is indistinguishable from a 2-parameter OU-optimal formula on this problem, because

LSTM's long-range memory is the wrong inductive bias for mean-reverting dynamics.

6.2 Future Work

- **Larger universe**: Expand to 100+ stocks with sector constraints to find multiple stable cointegrated pairs with stronger economic rationale.
- **GARCH volatility integration**: Use GARCH(1,1) conditional volatility to dynamically widen entry thresholds during high-volatility regimes, reducing false entries during turbulent periods.
- **Kalman innovation diagnostics**: Apply Ljung-Box tests to filter innovations to formally verify that Q and R are correctly specified.
- **Hybrid OU-LSTM**: Use OU-optimal forecast as the primary signal and LSTM residuals as a secondary filter, focusing the neural network on the nonlinear component that the OU model cannot capture.
- **Out-of-sample validation**: Reserve 2022–2024 data as a true holdout set, training all models on 2014–2021 only, to get an unbiased performance estimate.
- **Real-time deployment**: Integrate with live market data for intraday signal generation with position sizing proportional to cointegration p-value and current OU half-life.

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